

PROJECT BOB • TEAM APEX

AI-Based Crime Scene Evidence Prioritization

Case Study: The 2019 Hyderabad Veterinarian Murder
Investigation



The 2019 Catalyst

A massive data sweep resulting in investigator cognitive overload.

3,000+ **200+**

Scene photos

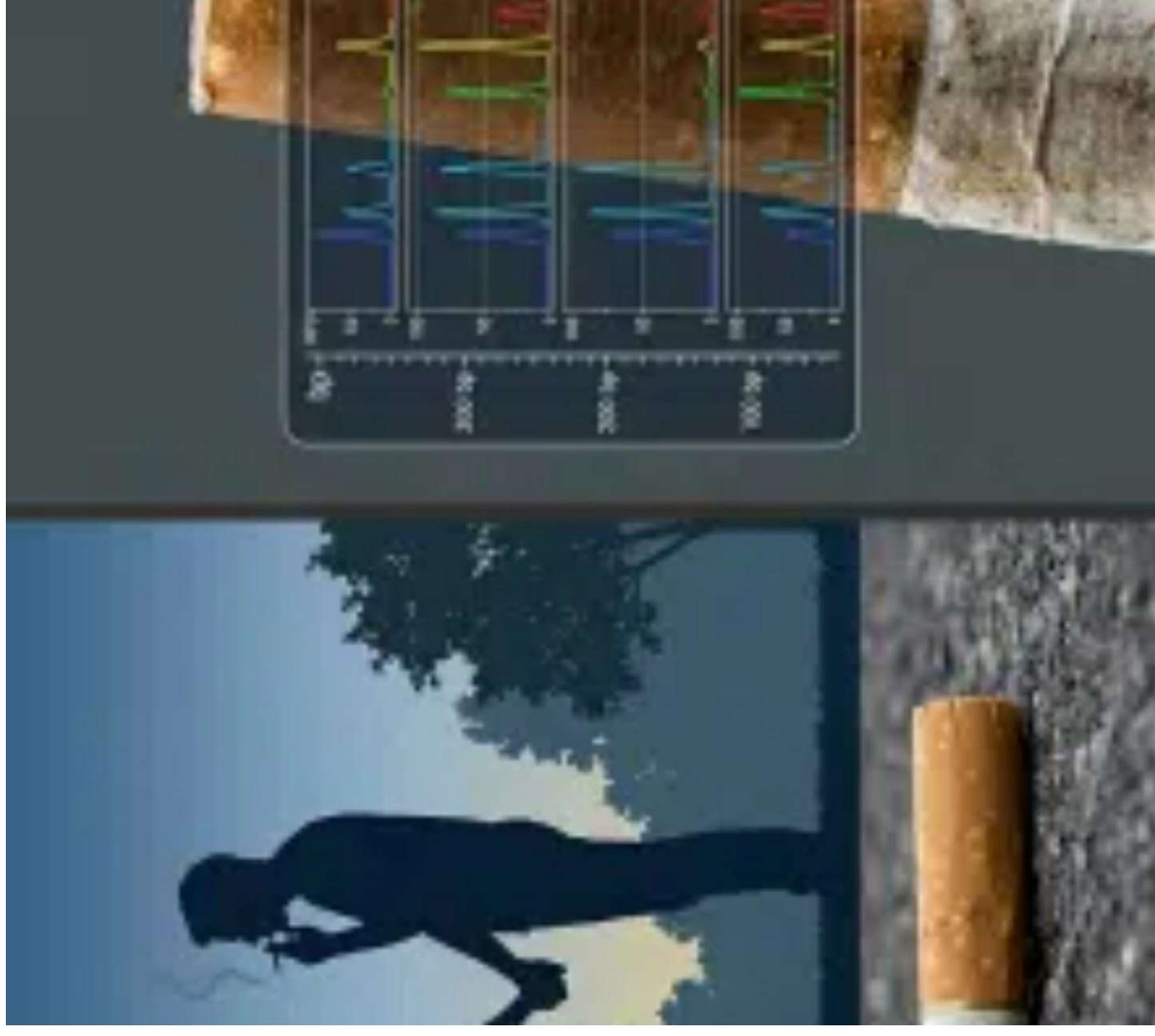
Physical items

| Zero automated triage protocols active during intake.

THE VULNERABILITY

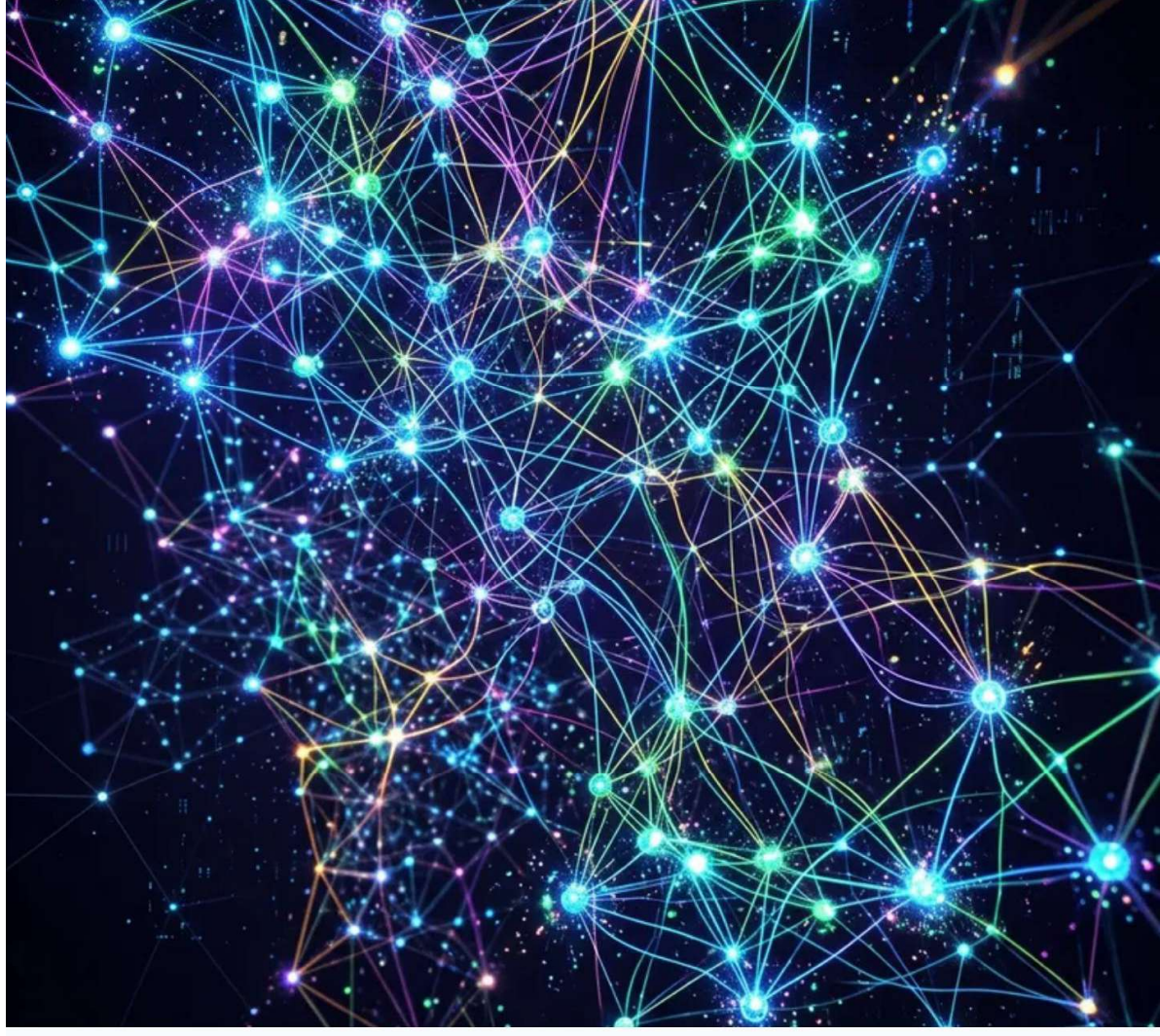
The Priority Paradox

- **The Near-Miss:** A crucial DNA-bearing cigarette butt was nearly overlooked during the manual sweep.
- **Indiscriminate Intake:** All items submitted without tiered priority ranking.
- **FSL Backlog:** Weeks of delay stall the most critical investigation phase.



THE SOLUTION

Introducing Bob: Evidence Triage Assistant



The Triage Workflow

1

Ingest

Investigator lists crime scene items.



2

Classify

Categorize by evidence modality.



3

Score

Assign probative priority value.



4

Route

Generate ranked FSL schedule.

Priority Matrix



Tier 1: Urgent

Biological materials and DNA.



Tier 2: Standard

Digital devices and trace evidence.



Tier 3: Deferred

Circumstantial context items.



OPERATIONAL IMPACT

Ranked FSL Routing

High-probative items immediately bypass routine queues.

✓ **Zero Misses:** Critical evidence identified instantly.

✓ **Turnaround:** Analysis reduced from weeks to days.

Team Apex

Transforming Forensic Intelligence with Bob.



Image Sources



<https://terratape.com/assets/components/phpthumbof/cache/bannerguard-police-night-scene.aa0eaca73e23307f31f4891851d42f24.jpg>

Source: terratape.com



<https://simplyforensic.com/wp-content/uploads/2026/03/dna-transfer-persistence-cigarette-butt-forensic-330x220.webp>

Source: simplyforensic.com



https://media.easy-peasy.ai/6c7a9112-5a43-46c2-9ebe-1cade0e9f155/fc5189a8-de88-44c9-bcf4-6933a2592410_medium.webp

Source: [easy-peasy.ai](https://media.easy-peasy.ai)



<https://cdn.labmanager.com/assets/image/54722/forensic-analytical-chemistry-s.webp>

Source: www.labmanager.com